Teensy, 5V Logic

File: Teensy, 5V Logic.kicad_sch

outputs

File: outputs.kicad_sch

Power

File: power.kicad_sch

UART

File: uart.kicad_sch

Spindle Control

File: Spindle Control_sch.kicad_sch

isolated input

File: isolated input_sch.kicad_sch

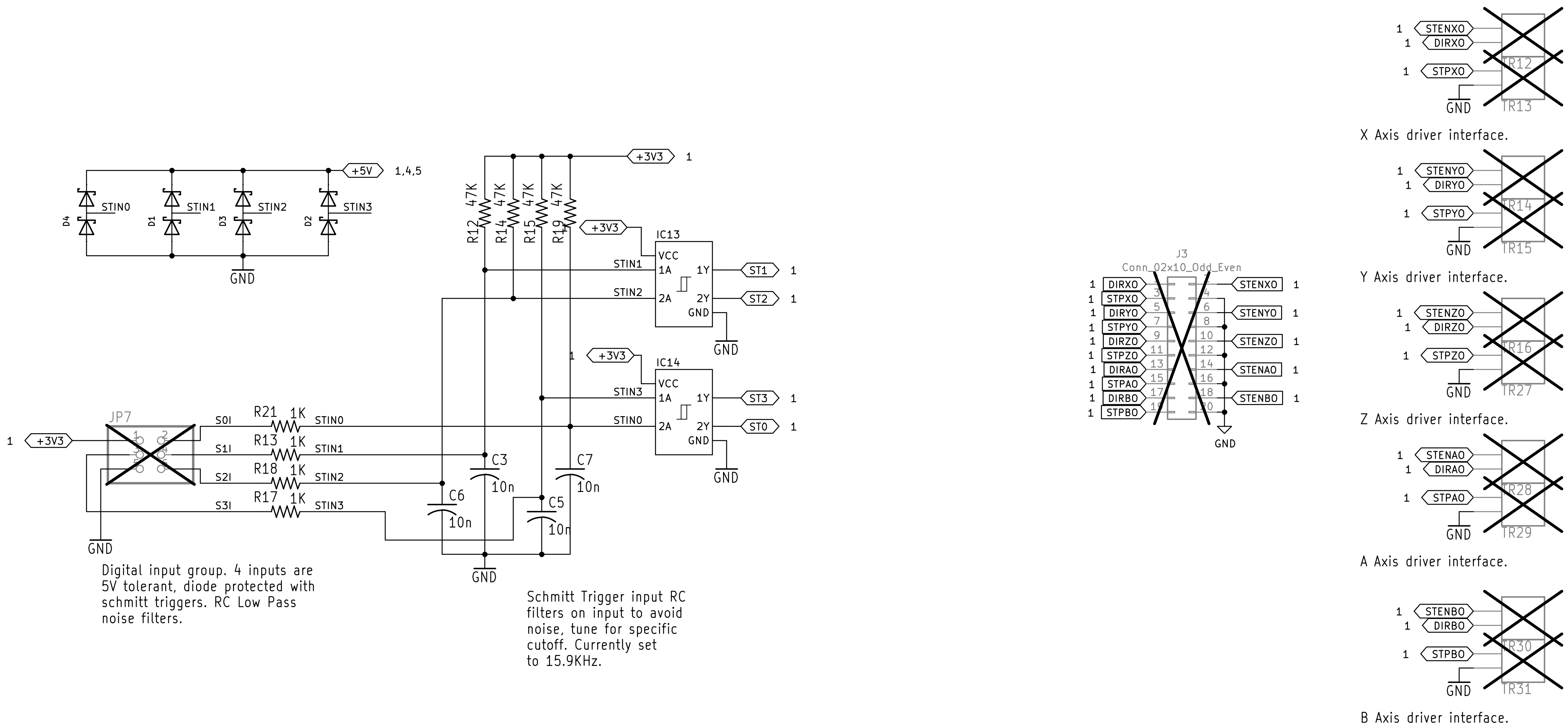
Brookwood Design

Sheet: /

File: V300.kicad_sch

Title: T41U5XBB

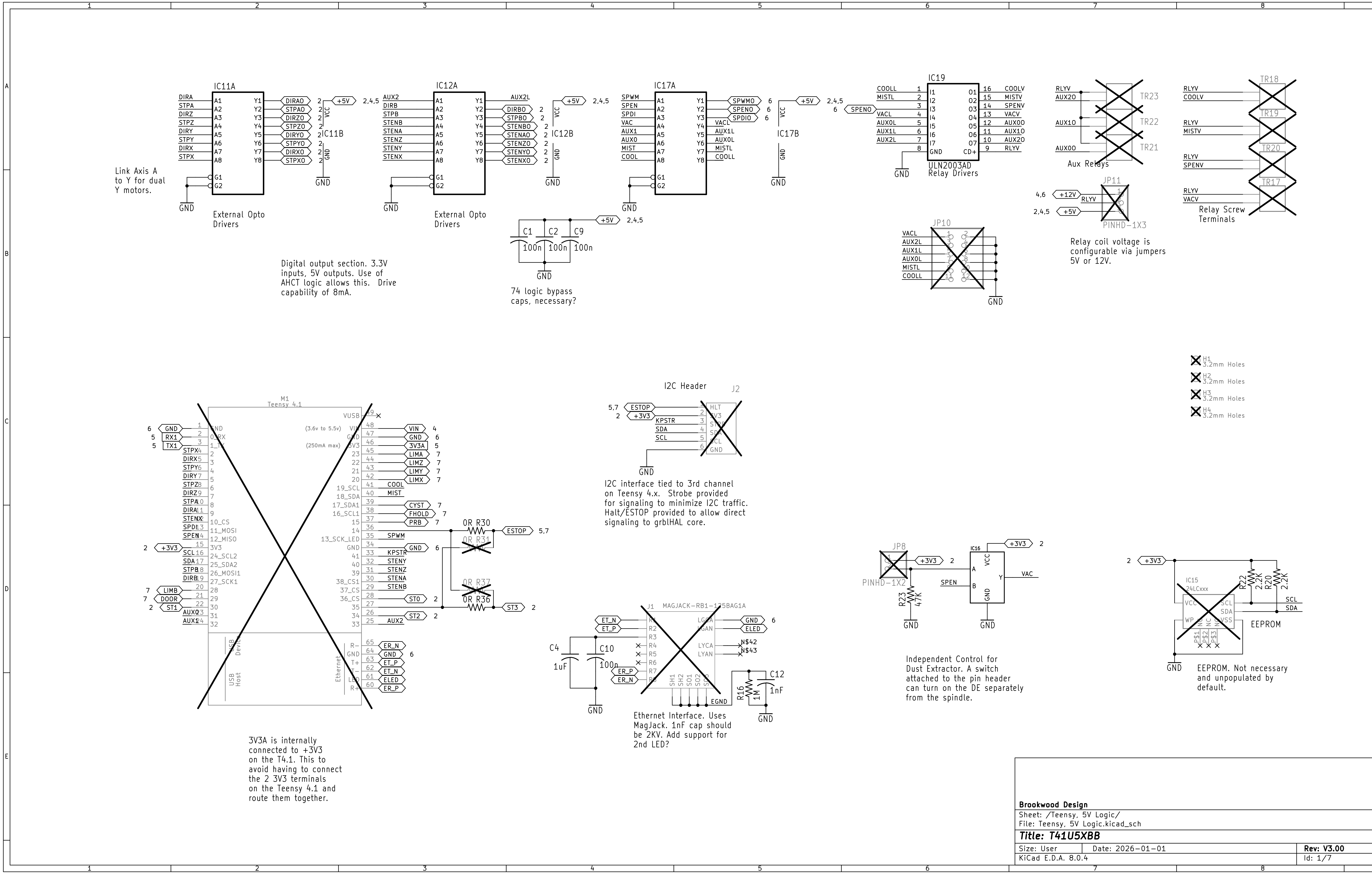
Size: User	Date: 2026-01-01	**Rev: V3.00**
KiCad E.D.A. 8.0.4		Id: 1/7



Brookwood Design
Sheet: /outputs/
File: outputs.kicad_sch

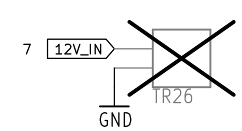
Title: T41U5XBB

Size: User | Date: 2026-01-01 | Rev: V3.00
KiCad E.D.A. 8.0.4 | Id: 1/7

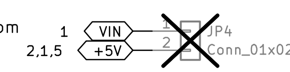


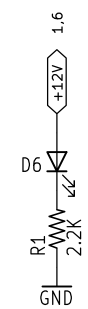
IC11A
External Opto Drivers
Link Axis A to Y for dual Y motors.
IC12A
External Opto Drivers
IC17A
IC19
ULN2003AD Relay Drivers
Aux Relays
TR23
TR22
TR21
TR18
TR19
TR20
TR17
Relay Screw Terminals
JP11
PINHD-1X3
Relay coil voltage is configurable via jumpers 5V or 12V.
JP10
Digital output section. 3.3V inputs, 5V outputs. Use of AHCT logic allows this. Drive capability of 8mA.
74 logic bypass caps, necessary?
H1 3.2mm Holes
H2 3.2mm Holes
H3 3.2mm Holes
H4 3.2mm Holes
M1
Teensy 4.1
I2C Header
J2
I2C interface tied to 3rd channel on Teensy 4.x. Strobe provided for signaling to minimize I2C traffic. Halt/ESTOP provided to allow direct signaling to grblHAL core.
JP8
PINHD-1X2
IC16
Independent Control for Dust Extractor. A switch attached to the pin header can turn on the DE separately from the spindle.
IC15
24LCxxx
EEPROM
EEPROM. Not necessary and unpopulated by default.
J1
MAGJACK-RB1-125BAG1A
Ethernet Interface. Uses MagJack. 1nF cap should be 2KV. Add support for 2nd LED?
3V3A is internally connected to +3V3 on the T4.1. This to avoid having to connect the 2 3V3 terminals on the Teensy 4.1 and route them together.
Brookwood Design
Sheet: /Teensy, 5V Logic/
File: Teensy, 5V Logic.kicad_sch
Title: T41U5XBB
Size: User
Date: 2026-01-01
Rev: V3.00
KiCad E.D.A. 8.0.4
Id: 1/7

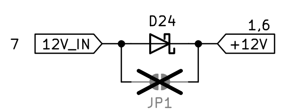
12V Input. Required for for
relays and spindle 0-10V.
Used to generate +5V for
digital logic.



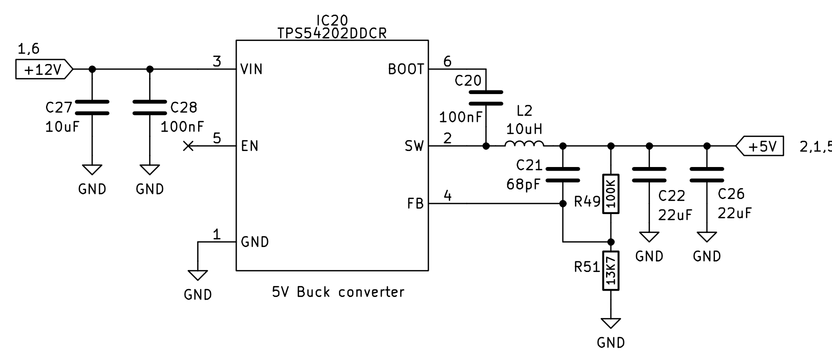
Short to power the Teensy from
internally generated +5V.







Reverse Polarity Protection
Jumper to allow removal if
it causes problems.



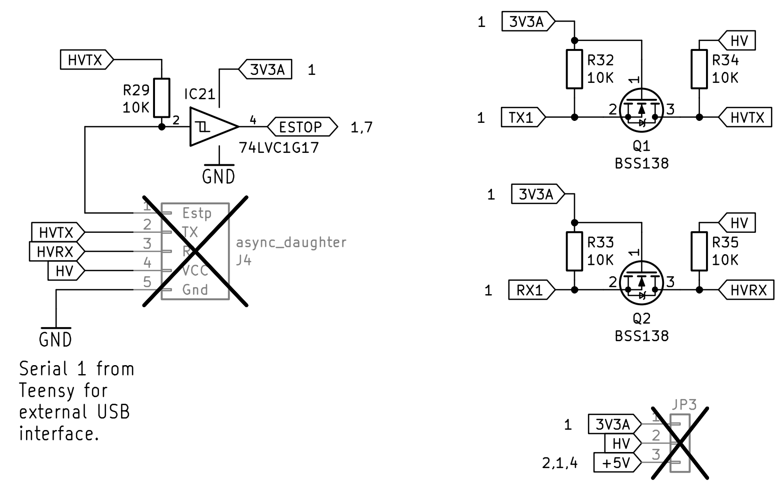
5V Buck converter

12V vs 24V

Support either voltage as
input. Need to make sure
docs are clear on how it
works and user precautions.

Make sure:
- all LED and opto resistors are
  properly size with load/power calcs.
  Check LED illumination at
  both voltages.
- Max voltage on all caps.
- labeling is clear (call it 12
  but document 24V compat?)
- Test to 10% over-voltage.

Brookwood Design	
Sheet: /Power/	
File: power.kicad_sch	
Title: T41U5XBB	
Size: User	Date: 2026-01-01
Rev: V3.00	
KiCad E.D.A. 8.0.4	Id: 1/7



Serial 1 from
Teensy for
external USB
interface.

Select Translation
Voltage

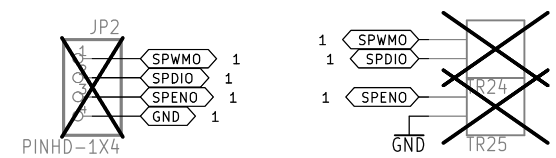
Brookwood Design

Sheet: /UART/

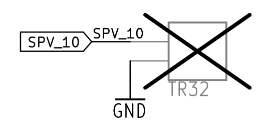
File: uart.kicad_sch

Title: T41U5XBB

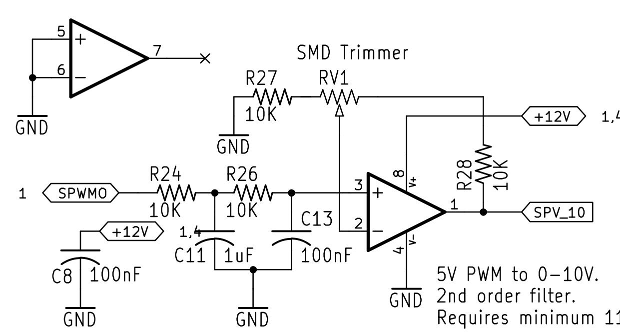
Size: User	Date: 2026-01-01	**Rev: V3.00**
KiCad E.D.A. 8.0.4		Id: 1/7



Spindle Control
interfaces. PWM,
Direction, Enable
and 0-10V analog.



Spindle Control. Supports direct
PWM and on/off (via Spindle enable)
as well as 0-10V control. 10V output
is adjustable via trim pot. Consider
replacing LM358 with smaller
opamp, single channel (JLC LM358
is Basic part and lowest cost Op
Amp).

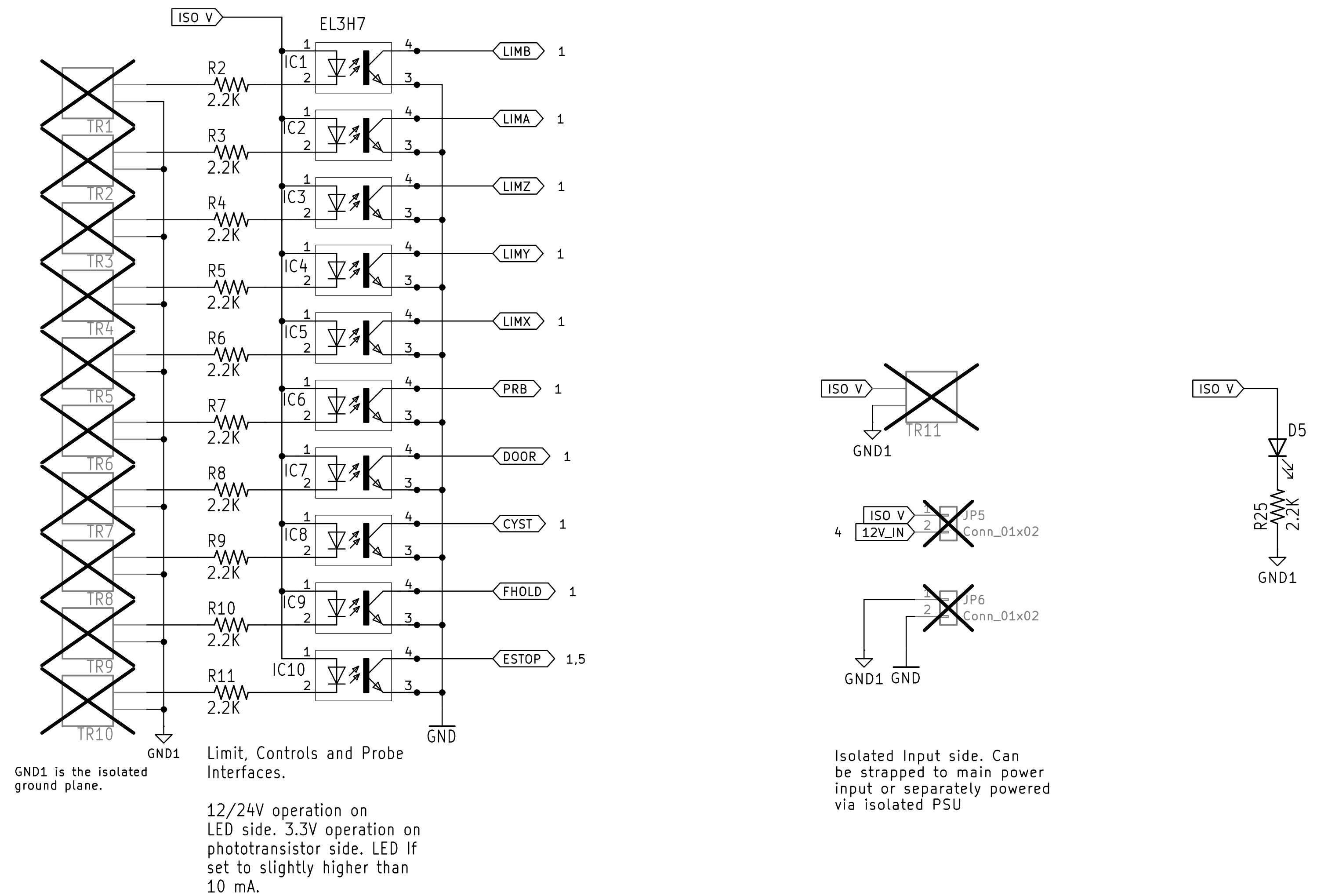


5V PWM to 0-10V.
2nd order filter.
Requires minimum 11.5V.

Brookwood Design
Sheet: /Spindle Control/
File: Spindle Control_sch.kicad_sch

Title: T41U5XBB

Size: User	Date: 2026-01-01	**Rev: V3.00**
KiCad E.D.A. 8.0.4		Id: 1/7



Brookwood Design

Sheet: /isolated input/

File: isolated input_sch.kicad_sch

Title: T41U5XBB

Size: User	Date: 2026-01-01	**Rev: V3.00**
KiCad E.D.A. 8.0.4		Id: 1/7